VESPAW

Theory and methods

Variational electronic structure with projector augmented waves

Mohsen Sotoudeh

Version 1.0.2

Contents

1 Introduction **1**
- 1.1 Why another look at an old problem . . . 1
- 1.2 VESPAW and CP-PAW . . . 1
- 1.3 How to read this book . . . 2

2 Density-functional theory **3**
- 2.1 Introduction . . . 3
- 2.2 The many-electron problem . . . 3
- 2.3 Hohenberg–Kohn theorems . . . 4
- 2.4 Kohn–Sham equations . . . 5
- 2.5 The exchange-correlation energy . . . 5
- 2.6 Exchange-correlation functionals . . . 6
- 2.7 Spin . . . 8
- 2.8 Finite temperature . . . 8

3 Plane waves and Brillouin-zone sampling **10**
- 3.1 Introduction . . . 10
- 3.2 Bloch states and plane waves . . . 10
- 3.3 Cutoffs and FFT grids . . . 10
- 3.4 k-point sampling . . . 11
- 3.5 The Γ point . . . 11

4 The projector augmented-wave method **12**
- 4.1 Introduction . . . 12
- 4.2 The PAW transformation . . . 12
- 4.3 Compensation charges, core and local potential . . . 13
- 4.4 Setup construction . . . 14
- 4.5 Quality checks . . . 14
- 4.6 The setup library . . . 15
- 4.7 Core-level spectra . . . 15

5 Local hybrid functionals **17**
- 5.1 Introduction . . . 17
- 5.2 Hybrid functionals . . . 17
- 5.3 Local orbitals . . . 18
- 5.4 Projection onto the local orbitals . . . 18
- 5.5 PBE0r . . . 18
- 5.6 PBE-SL0r . . . 19
- 5.7 SCAN0r . . . 19

6 Self-consistency: the variational optimizer **21**
6.1 Introduction . . . 21
6.2 The problem with density mixing . . . 21
6.3 Free energy as a function of the Hamiltonian . . . 22
6.4 The gauge direction . . . 22
6.5 Line search and acceptance . . . 22
6.6 Route 2: direct joint minimization . . . 23
6.7 The consistency test . . . 23

7 Eigensolvers and dynamics **24**
7.1 Introduction . . . 24
7.2 Block Davidson . . . 24
7.3 RMM-DIIS . . . 25
7.4 Conjugate gradients . . . 25
7.5 Car–Parrinello dynamics with friction . . . 25

8 Forces, stress and molecular dynamics **26**
8.1 Introduction . . . 26
8.2 Forces . . . 26
8.3 Stress . . . 27
8.4 Structure relaxation and molecular dynamics . . . 27

9 Structure of the code **28**
9.1 Introduction . . . 28
9.2 Source files . . . 28
9.3 One SCF step . . . 29
9.4 Input . . . 29
9.5 Adding a functional . . . 29
9.6 Adding an optimizer . . . 29
9.7 Regression suite . . . 29

Chapter 1

Introduction

1.1 Why another look at an old problem 1
1.2 VESPAW and CP-PAW 1
1.3 How to read this book 2

1.1 Why another look at an old problem

A first-principles calculation of a material ends with a number, a total energy, a band gap, a magnetic moment. Whether this number means anything depends on a long chain of steps: the density functional, the basis set, the treatment of the atomic cores, the sampling of the Brillouin zone, and finally the iteration that makes the electronic structure self-consistent. Most of these steps are controlled by a few numbers that can be converged. The last one is different. A self-consistency cycle that has stalled, or that has settled in a metastable state, may report convergence just like one that has found the ground state, and nothing in the output tells the difference.

For an experienced user who looks at every calculation, this is a nuisance. For the large sets of calculations that are now used to search for new materials for batteries, catalysts and other energy technologies, it is a real problem, because nobody looks at every calculation any more. The materials that matter most, transition-metal oxides with magnetic order, mixed valence and polarons, are exactly the ones where self-consistency is most delicate.

VESPAW starts from this observation. Its central idea is to treat the effective Kohn–Sham Hamiltonian, and not the density, as the variable of the self-consistency problem, and to evaluate the energy for the actual state that each trial Hamiltonian produces. Every energy is then an upper bound of the ground-state energy, and the iteration accepts a step only if this bound does not increase. In this book I describe the physics and the numerical methods on which VESPAW rests, with the emphasis on how the Kohn–Sham equations are solved.

1.2 VESPAW and CP-PAW

VESPAW is a plane-wave code based on the projector augmented-wave (PAW) method [1]. It is a fork of CP-PAW, the Car–Parrinello PAW code written by Peter E. Blöchl at Clausthal University of Technology. The PAW formalism, the construction of the setups, the Car–Parrinello dynamics, the local orbitals and the local hybrid functional PBE0r, and the input format all come from CP-PAW. VESPAW adds

- a *variational* self-consistency optimizer, whose energies are upper bounds of the ground-state energy in every iteration (Chapter 6);
- iterative eigensolvers (block Davidson, RMM-DIIS, conjugate gradients) next to the damped dynamics of CP-PAW (Chapter 7);
- functionals checked against libxc and VASP, the SCAN family (SCAN-L, and SCAN and r^{2}SCAN with the orbital kinetic-energy density), and the local hybrids PBE-SL0r and SCAN0r, which build on PBE0r (Chapters 2 and 5);
- quality checks for PAW setups, an audited setup library (Chapter 4) and a regression suite (Chapter 9).

License. CP-PAW is distributed under the GNU General Public License, version 3 (GPL-3.0). VESPAW is derived from CP-PAW and is distributed under the same license. The older notice "(C) Copyright Clausthal University of Technology, licensed materials" in the headers of the inherited source files predates the GPL release.

1.3 How to read this book

Chapters 2 to 4 summarize the physics that VESPAW implements: density-functional theory, plane waves and the PAW method, including core-level spectra. Chapter 5 treats hybrid and local hybrid functionals. Chapters 6 and 7 describe how the Kohn–Sham equations are solved, which is where VESPAW differs most from other codes. Chapter 8 covers forces, stress and molecular dynamics, and Chapter 9 the structure of the program. The hands-on tutorial (`doc/tutorial.pdf`) puts all of this into practice.

Input keywords are written in **`bold typewriter`**; their complete description is in the input manual. Hartree atomic units are used throughout unless stated otherwise, while cutoff energies are given in Rydberg, as in the input. The program reports its results in eV, Å and GPa.

Pitfalls. Many sections end with a short list of pitfalls. They record errors that were actually found in the code and fixed. I have kept them, because they show which quantities are delicate and how they are tested.

Chapter 2

Density-functional theory

2.1 Introduction . . . 3
2.2 The many-electron problem . . . 3
2.3 Hohenberg–Kohn theorems . . . 4
2.4 Kohn–Sham equations . . . 5
2.5 The exchange-correlation energy . . . 5
2.6 Exchange-correlation functionals . . . 6
2.7 Spin . . . 8
2.8 Finite temperature . . . 8

2.1 Introduction

The electrons of a solid interact with each other, and their wave function depends on the coordinates of all of them at once. Already for a few dozen electrons such a wave function cannot even be stored, let alone calculated. Density-functional theory removes this obstacle in two steps. Hohenberg and Kohn showed that the ground-state energy is a functional of the electron density alone, a function of only three coordinates. Kohn and Sham then expressed this functional through a fictitious system of non-interacting electrons that has the same density. All the many-body physics is collected in one term, the exchange-correlation energy, and the art of density-functional theory lies in its approximation. This chapter summarizes the theory and the functionals implemented in VESPAW, with particular attention to how their parameters were checked against independent implementations.

2.2 The many-electron problem

Hamiltonian. A system of nuclei at positions $\boldsymbol{R}_1, \ldots, \boldsymbol{R}_M \equiv \boldsymbol{R}$ with masses M_I and charges Z_I, and of N electrons with coordinates $\boldsymbol{x}_i = (\boldsymbol{r}_i, s_i)$, combining position and spin, is described by the Hamiltonian

$$\hat{H} = \underbrace{-\sum_I \frac{\nabla^2_{\boldsymbol{R}_I}}{2M_I}}_{\hat{T}_\mathrm{N}} \underbrace{-\sum_i \tfrac{1}{2}\nabla^2_{\boldsymbol{r}_i}}_{\hat{T}_\mathrm{e}} + \underbrace{\sum_{i<j} \frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}}_{\hat{V}_\mathrm{ee}} \underbrace{-\sum_{i,I} \frac{Z_I}{|\boldsymbol{r}_i - \boldsymbol{R}_I|}}_{\hat{V}_\mathrm{eN}} + \underbrace{\sum_{I<J} \frac{Z_I Z_J}{|\boldsymbol{R}_I - \boldsymbol{R}_J|}}_{\hat{V}_\mathrm{NN}} . \tag{2.1}$$

In principle, every property of molecules and solids follows from the eigenstates of this operator. In practice, the number of electronic configurations grows exponentially with the number of electrons, and the problem must be simplified before it can be solved.

Born–Oppenheimer approximation. Nuclei are more than three orders of magnitude heavier than electrons. Born and Oppenheimer [2, 3] therefore first solve the electronic problem for fixed nuclei,

$$\hat{H}_\mathrm{e}(\boldsymbol{R})\,\Phi_k(\boldsymbol{x};\boldsymbol{R}) = E_k(\boldsymbol{R})\,\Phi_k(\boldsymbol{x};\boldsymbol{R}), \qquad \hat{H}_\mathrm{e} = \hat{T}_\mathrm{e} + \hat{V}_\mathrm{ee} + \hat{V}_\mathrm{eN} + \hat{V}_\mathrm{NN}, \tag{2.2}$$

and expand the full wave function as $\Psi(\boldsymbol{x},\boldsymbol{R},t) = \sum_k \Phi_k(\boldsymbol{x};\boldsymbol{R})\chi_k(\boldsymbol{R},t)$. This expansion is exact. Inserting it into the time-dependent Schrödinger equation gives equations of motion for the nuclear wave functions χ_k that are coupled through matrix elements of $\nabla_{\boldsymbol{R}}$ between different electronic states. The Born–Oppenheimer approximation neglects these non-adiabatic couplings: the nuclei then move on a single potential-energy surface $E_k(\boldsymbol{R})$. The approximation fails where two surfaces come close, within the thermal energy $k_B T$ or within the energy of the nuclear motion. Treating the nuclei moreover as classical particles gives Newton's equations

$$M_I \ddot{\boldsymbol{R}}_I = -\nabla_{\boldsymbol{R}_I} E_0(\boldsymbol{R}), \tag{2.3}$$

which is the basis of molecular dynamics (Chapter 8). By the Hellmann–Feynman theorem [4] the force is the expectation value $-\langle\Phi_0|\nabla_{\boldsymbol{R}_I}\hat{H}_\mathrm{e}|\Phi_0\rangle$. What remains is to find the electronic ground state Φ_0 and its energy $E_0(\boldsymbol{R})$.

Variational principle. For any normalizable, antisymmetric trial wave function Φ,

$$E[\Phi] = \frac{\langle\Phi|\hat{H}_\mathrm{e}|\Phi\rangle}{\langle\Phi|\Phi\rangle} \geq E_0, \tag{2.4}$$

with equality only for the ground state. Wave-function methods restrict Φ to a subspace: Hartree–Fock theory to a single Slater determinant, configuration interaction to linear combinations of determinants [5]. Their accuracy can be improved systematically by enlarging the subspace, but the cost grows so steeply that only small systems can be treated. Density-functional theory avoids this cost by replacing the many-electron wave function with the density. The upper-bound property of Eq. (2.4) carries over to it, and it is the foundation of the variational optimizer of Chapter 6.

2.3 Hohenberg–Kohn theorems

The electron density

$$n(\boldsymbol{r}) = N \sum_s \int d^4x_2 \cdots \int d^4x_N\, |\Phi(\boldsymbol{r}s, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N)|^2, \qquad \int d^4x := \sum_s \int d^3r, \tag{2.5}$$

depends on three coordinates only. Hohenberg and Kohn [6] showed that it nevertheless determines all ground-state properties.

First theorem. The external potential is determined, up to a constant, by the ground-state density. Suppose that two potentials v_1 and v_2, which differ by more than a constant, lead to non-degenerate ground states Φ_1 and Φ_2 with the same density n. Because Φ_2 is not the ground state of $\hat{H}_1$,

$$E_1 < \langle\Phi_2|\hat{H}_1|\Phi_2\rangle = \langle\Phi_2|\hat{H}_2|\Phi_2\rangle + \langle\Phi_2|\hat{H}_1 - \hat{H}_2|\Phi_2\rangle = E_2 + \int d^3r\, n(\boldsymbol{r})\big[v_1(\boldsymbol{r}) - v_2(\boldsymbol{r})\big]. \tag{2.6}$$

Exchanging the indices gives $E_2 < E_1 + \int d^3r\, n\,(v_2 - v_1)$, and the sum of both inequalities gives the contradiction $E_1 + E_2 < E_1 + E_2$. Since v fixes the Hamiltonian, the density fixes the ground state and all properties derived from it.

Second theorem. The energy can be written as

$$E_v[n] = F[n] + \int d^3r\, v(\boldsymbol{r})\, n(\boldsymbol{r}), \qquad F[n] = \min_{\Phi\to n} \langle \Phi | \hat{T}_\mathrm{e} + \hat{V}_\mathrm{ee} | \Phi \rangle, \tag{2.7}$$

where F is a universal functional, the same for all systems, and the minimum runs over all wave functions with density n (constrained search of Levy [7], which also removes the restriction to non-degenerate ground states). By Eq. (2.4), $E_v[n]$ is minimized by the ground-state density, and its minimum is the ground-state energy. If F were known, the many-electron problem would be solved for all external potentials at once. The theorems hold for the ground state only; excited states require extensions such as time-dependent density-functional theory.

2.4 Kohn–Sham equations

The kinetic energy is the largest part of $F[n]$ and the hardest to express directly by the density. Kohn and Sham [8] therefore introduce a fictitious system of non-interacting electrons that has the same density as the interacting one. Its kinetic energy T_s is calculated exactly from its orbitals, and only the remainder is approximated. With non-interacting orbitals ψ_n with occupations f_n,

$$E[n] = \sum_n f_n \langle \psi_n | -\tfrac{1}{2}\nabla^2 | \psi_n \rangle + \int d^3r\, v_\mathrm{ext} n + \frac{1}{2} \iint d^3r\, d^3r' \, \frac{n(\boldsymbol{r}) n(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|} + E_\mathrm{xc}[n], \qquad n(\boldsymbol{r}) = \sum_n f_n |\psi_n(\boldsymbol{r})|^2, \tag{2.8}$$

the terms are the kinetic energy of the non-interacting electrons, the interaction with the external potential of the nuclei, the Hartree energy, and the exchange-correlation energy, which contains everything else. The nuclear repulsion V_NN is a constant for fixed nuclei and is added to E; it matters for forces and stress. Minimizing with orthonormal orbitals gives the Kohn–Sham equations

$$\Big[-\tfrac{1}{2}\nabla^2 + v_\mathrm{eff}(\boldsymbol{r}) \Big] \psi_n = \epsilon_n \psi_n, \qquad v_\mathrm{eff} = v_\mathrm{ext} + v_\mathrm{H}[n] + \frac{\delta E_\mathrm{xc}}{\delta n}. \tag{2.9}$$

Because v_eff depends on the density, the equations must be solved self-consistently (Chapter 6). A self-consistent solution is a stationary point of E only if the potential is the functional derivative of the energy. This condition is easy to state and easy to violate in an implementation; VESPAW tests it routinely (Sec. 6.7).

2.5 The exchange-correlation energy

Exchange-correlation hole. If the electrons were uncorrelated, the probability of finding one electron at $\boldsymbol{r}$ and another at $\boldsymbol{r}'$ would be the product $n(\boldsymbol{r})n(\boldsymbol{r}')$, and their interaction would be the Hartree energy. Real electrons avoid each other, because of the Pauli principle (exchange) and because of their Coulomb repulsion (correlation). The pair density is therefore written as

$$n^{(2)}(\boldsymbol{r}, \boldsymbol{r}') = n(\boldsymbol{r}) \big[n(\boldsymbol{r}') + h(\boldsymbol{r}, \boldsymbol{r}') \big], \tag{2.10}$$

where the exchange-correlation hole $h(\boldsymbol{r}, \boldsymbol{r}')$ describes the depletion of density around an electron at $\boldsymbol{r}$. It obeys the sum rule

$$\int d^3r'\, h(\boldsymbol{r}, \boldsymbol{r}') = -1 : \tag{2.11}$$

an electron sees exactly $N - 1$ others. The electron–electron interaction becomes the Hartree energy plus the interaction of each electron with its own hole,

$$V_\mathrm{ee} = E_\mathrm{H} + U_\mathrm{xc}, \qquad U_\mathrm{xc} = \frac{1}{2} \int d^3r\, n(\boldsymbol{r}) \int d^3r' \, \frac{h(\boldsymbol{r}, \boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|}. \tag{2.12}$$

U_xc is not yet E_xc. The exchange-correlation energy also contains the difference $T - T_s$ between the kinetic energy of the interacting electrons and that of the Kohn–Sham system.

Adiabatic connection. Both contributions are combined by switching on the interaction, $\hat{V}_{ee} \to \lambda \hat{V}_{ee}$, while an external potential keeps the density fixed [9–11]. At $\lambda = 0$ one has the Kohn–Sham system, at $\lambda = 1$ the real one. Integrating over λ gives

$$E_{xc} = \int_0^1 d\lambda\, U_{xc}^{\lambda} = \frac{1}{2} \int d^3r\, n(\boldsymbol{r}) \int d^3r'\, \frac{\bar{h}(\boldsymbol{r}, \boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|}, \qquad \bar{h} = \int_0^1 d\lambda\, h^{\lambda}, \tag{2.13}$$

so the exchange-correlation energy is the electrostatic energy of the electrons with their coupling-constant-averaged hole; the kinetic part is absorbed in $\bar{h}$. At $\lambda = 0$ the hole is the exchange hole of the Kohn–Sham determinant, and U_{xc}^0 is the exact (Hartree–Fock-like) exchange energy E_x evaluated with the Kohn–Sham orbitals. Eq. (2.13) is the starting point of the hybrid functionals (Chapter 5). The sum rule explains why simple approximations of $\bar{h}$ work better than one might expect: only the spherical average of the hole enters the energy, and its normalization is fixed.

Self-interaction. For a single electron, with density $n_\uparrow = n$, $n_\downarrow = 0$, there is no electron–electron interaction. The exact functional must then satisfy

$$E_{xc}[n, 0] = -E_H[n], \qquad E_c[n, 0] = 0 : \tag{2.14}$$

exchange cancels the spurious interaction of the electron with itself in the Hartree term. Semilocal functionals violate this condition. The self-interaction error $E_H[n] + E_{xc}[n, 0]$ delocalizes electrons, in particular the d and f electrons of transition-metal and rare-earth compounds, and reduces band gaps. Exact exchange is free of self-interaction; this is the motivation for the hybrid functionals of Chapter 5.

2.6 Exchange-correlation functionals

LDA. The local-density approximation, proposed by Kohn and Sham [8], takes at each point the exchange-correlation energy per electron ϵ_{xc}^{hom} of the homogeneous electron gas with the local density,

$$E_{xc}^{LDA} = \int d^3r\, n(\boldsymbol{r})\, \epsilon_{xc}^{hom}\big(n(\boldsymbol{r})\big). \tag{2.15}$$

Exchange is known analytically; correlation is parametrized from the quantum Monte Carlo data of Ceperley and Alder [12] (PZ [13], PW92 [14]). The LDA works well for many solids, but it overbinds, severely so for molecules.

GGA. Generalized gradient approximations [15] multiply the LDA energy density by an enhancement factor that depends on the density and its gradient,

$$E_{xc}^{GGA} = \int d^3r\, n\, \epsilon_x^{hom}(n)\, F_{xc}(r_s, s), \qquad s = \frac{|\nabla n|}{2k_F n}, \qquad k_F = (3\pi^2 n)^{1/3}, \tag{2.16}$$

with the Seitz radius $r_s = (3/4\pi n)^{1/3}$ and the reduced gradient s. GGAs are not simply the next order of a gradient expansion, which is poor for real densities; they are constructed to satisfy exact conditions such as the sum rule of the hole. They correct most of the overbinding of the LDA and made density-functional theory useful in chemistry. The default in VESPAW is PBE [16].

Selection. All functionals are selected by **`!CONTROL!DFT TYPE`**, as a number or a name, e.g. **`TYPE='PBE'`**; the VASP **`GGA`** tags are accepted as names. Table 2.1 lists them.

Parameters. The constants follow the original publications and agree with libxc and VASP: $\beta = 0.06672455060314922$ in PBE correlation and in $\mu = \beta\pi^2/3$ of PBE exchange (before October 2026 VESPAW used $\beta = \nu C_c(0) = 0.0667263$, kept by **`TYPE=-10`** together with the pre-2024 reduced gradient, t^2 too small by a factor 0.62); PW91 keeps $\beta = \nu C_c(0)$, as in VASP; PBEsol uses $\mu = 10/81$ and $\beta = 0.046$; SCAN-L uses the optimized Perdew–Constantin parameters $a = 1.784720$, $b = 0.258304$ of Ref. [26]; the spin interpolation of SCAN correlation uses $G_c(\zeta) = [1 - 2.363(d_x(\zeta) - 1)](1 - \zeta^{12})$, with 2.363 as in VASP and libxc (the SCAN supplement quotes 2.3631). Unit conversions use CODATA 2018 (1 Hartree = 27.211386 eV, 1 bohr = 0.529177211 Å). Plane-wave hybrids with full exact exchange (PBE0, HSE06, SCAN0, B3LYP) are not yet available.

Meta-GGAs: SCAN, r^{2}SCAN, SCAN-L and TSCAN. The SCAN functional [24] depends in addition on the kinetic energy density $\tau = \frac{1}{2}\sum_n f_n|\nabla\psi_n|^2$ through the iso-orbital indicator

$$\alpha = \frac{\tau - \tau_W}{\tau_{\text{unif}}}, \qquad \tau_W = \frac{|\nabla n|^2}{8n}, \qquad \tau_{\text{unif}} = \tfrac{3}{10}(3\pi^2)^{2/3} n^{5/3}. \tag{2.17}$$

VESPAW provides two routes:

- **SCAN-L** (**`TYPE='SCANL'`**), the deorbitalized SCAN of Mejía-Rodríguez and Trickey [26, 28], in which α is approximated from n, $|\nabla n|$ and $\nabla^2 n$. The functional depends on the Laplacian of the density, so its potential contains second-derivative terms (see below).
- **TSCAN** (**`TYPE='SCAN'`** with **`TSCANDELOC=T`**), the orbital-dependent SCAN with the true τ. The Hamiltonian then contains the operator $-\frac{1}{2}\nabla \cdot v_\tau \nabla$ with $v_\tau = \partial e_{\text{xc}}/\partial\tau$. In the PAW spheres the one-center τ is built from the partial waves, including its non-spherical parts.
- **r^{2}SCAN** (**`TYPE='R2SCAN'`** with **`TSCANDELOC=T`**), the regularized and restored SCAN of Furness *et al.* [25], with the same orbital τ and the same PAW treatment as TSCAN. It replaces α by the regularized $\bar\alpha = (\tau - \tau_W)/(\tau_{\text{unif}} + \eta\tau_W)$ with $\eta = 10^{-3}$, the switching function of SCAN by a polynomial for $0 \le \bar\alpha \le 2.5$, and adds the gradient-expansion corrections of exchange and correlation. The implementation follows the paper and its erratum and agrees with libxc to 10^{-14}. Band gaps change smoothly with the cutoff, unlike those of SCAN, which scatter by a few 0.01 eV.

PAW treatment of τ. The kinetic-energy density is divided like the density: the plane-wave part $\tilde\tau$ of the auxiliary wave functions plus the pseudized core kinetic-energy density $\tilde\tau_c$, and in each sphere the one-center terms $\tau^1 - \tilde\tau^1 = \sum_{ij} D_{ij}\left(\frac{1}{2}\nabla\phi_i \cdot \nabla\phi_j - \frac{1}{2}\nabla\tilde\phi_i \cdot \nabla\tilde\phi_j\right)$ plus the core parts. Two checks are useful: $\int \tilde\tau\, d^3r$ equals the kinetic energy of the auxiliary wave functions, and $\int(\tau^1 - \tilde\tau^1)\, d^3r = \sum_{ij} D_{ij}\Delta T_{ij}$. The setups of the meta-GGAs are constructed with PBE, as for VASP.

Functional derivatives. For a semilocal energy $E = \int e(n, |\nabla n|^2, \nabla^2 n)\, d^3r$ the potential is the full Euler–Lagrange derivative

$$v = \frac{\partial e}{\partial n} - \nabla \cdot \left(2\frac{\partial e}{\partial |\nabla n|^2}\nabla n\right) + \nabla^2 \frac{\partial e}{\partial \nabla^2 n}. \tag{2.18}$$

For GGAs the last term is absent; for SCAN-L it is essential. Where the energy carries a spatial weight (the double counting of the local hybrids, Sec. 5.6), the weight stands *inside* the derivatives. In the spherical one-center expansion the divergence becomes $r^{-2}(r^2 A)'$; in the non-spherical case the weak form is used, projecting $\partial e/\partial\nabla n$ onto the radial and tangential gradients of each spherical harmonic.

Pitfall. Before September 2026, the one-center SCAN-L and TSCAN potentials kept only the first term of Eq. (2.18). Self-consistent solutions were then not stationary points of the energy: the variational optimizer found lower energies without reaching self-consistency, and for TSCAN it did not converge at all. The consistency test of Sec. 6.7 exposed both.

Pitfall (October 2026). Three errors made all meta-GGA band gaps too small by about 0.3 eV, e.g. 4.93 instead of 5.24 eV for SCAN MgO: the setups were built from an atom without exchange and correlation, the one-center τ lacked the factor $\frac{1}{2}$, and the pseudized core τ was missing in the plane-wave part. The energy and the Hamiltonian were consistent with each other, so the consistency tests passed. They were found with the two checks above and by comparison with VASP: SCAN and r^{2}SCAN gaps now agree within 0.06 eV (MgO 5.30 and 5.34 eV, VASP 5.24 and 5.30 eV; Si 0.949 and 0.919 eV, VASP 0.969 and 0.917 eV). For SCAN0r the double counting also subtracted the potential of the core τ, which made the Hamiltonian inconsistent with the energy and stalled the variational optimizer.

2.7 Spin

Collinear spin polarization (**`!STRUCTURE!OCCUPATIONS NSPIN=2`**) uses two sets of orbitals and the spin densities $n_\uparrow, n_\downarrow$; the code works internally with total and spin density (n, m). Non-collinear magnetism uses two-component spinors. For exchange, the spin-scaling relation

$$E_x[n_\uparrow, n_\downarrow] = \tfrac{1}{2}\big(E_x[2n_\uparrow] + E_x[2n_\downarrow]\big) \tag{2.19}$$

is exact and allows any spin-unpolarized exchange functional to be applied to a spin-polarized density. VESPAW uses it for the exchange of the SCAN family, on the grid and in the one-center terms. Correlation does not obey Eq. (2.19); SCAN correlation is evaluated for the spin densities with the spin interpolation of Ref. [24]: $\phi(\zeta)$ in the gradient term, $G_c(\zeta)$ in the single-orbital limit, and α divided by $d_s(\zeta) = [(1+\zeta)^{5/3} + (1-\zeta)^{5/3}]/2$ with the total τ. For SCAN-L, the spin-resolved kinetic-energy densities are deorbitalized separately, $\tau[n_\uparrow, n_\downarrow] = \frac{1}{2}(\tau[2n_\uparrow] + \tau[2n_\downarrow])$, as in libxc.

2.8 Finite temperature

For metals and for systems with near-degenerate states, VESPAW minimizes the Mermin free energy [29]

$$A = E[\{\psi_n\}, \{f_n\}] - TS[\{f_n\}], \qquad S = -k_B \sum_n \big[f_n \ln f_n + (1-f_n)\ln(1-f_n)\big], \tag{2.20}$$

with Fermi occupations $f_n = [1 + e^{(\epsilon_n - \mu)/k_B T}]^{-1}$ at fixed electron number. The electronic temperature is set in **`!CONTROL!MERMIN T[K]`**. It acts as a smearing parameter; the free energy A, not E, is the variational quantity.

Further reading

The original work on the subject of this chapter is found in Refs. [2, 6–8, 11, 16, 24, 29].

Table 2.1: Exchange-correlation functionals of VESPAW. "libxc" marks functionals whose energy density agrees with libxc 7.1.2 (and the VASP implementation) to $\sim 10^{-15}$ at every point of a grid in density, gradient, spin polarization and α (October 2026). The local hybrids are described in Chapter 5.

TYPE	name	functional	VASP	libxc
LDA				
1	**PZ**	Perdew–Zunger [13] of Ceperley–Alder [12]		✓
2	**PW92**	Perdew–Wang 92 [14]		✓
3		Slater exchange only		✓
4		X_α, $\alpha = 0.7$		
GGA				
10	**PBE**, **PE**	PBE [16]	**GGA=PE**	✓
11	**RPBE**, **RP**	RPBE [17]	**GGA=RP**	✓
12	**PW91**, **91**	PW91 exchange and correlation [18]	**GGA=91**	✓
13	**REVPBE**, **RE**	revPBE, $\kappa = 1.245$ [19]	**GGA=RE**	✓
14	**PBESOL**, **PS**	PBEsol [20]	**GGA=PS**	✓
5001	**BLYP**	B88 [21] + LYP [22]		✓
81		BP86: PZ + B88 + P86 [23]		✓
8		B88 + P86 on PW92 local correlation (non-standard)		
7, 71, 9		B88 exchange with PW92, PZ or PBE correlation		✓
6		B88 exchange only		✓
meta-GGA				
21	**SCAN**	SCAN, orbital τ [24] (with **TSCANDELOC=T**)	**METAGGA=SCAN**	✓
22	**R2SCAN**	r^{2}SCAN, orbital τ [25] (with **TSCANDELOC=T**)	**METAGGA=R2SCAN**	✓
20	**SCANL**	SCAN-L, deorbitalized [26, 27]	**METAGGA=SCANL**	✓
local hybrids (Chapter 5)				
	PBE0r	PBE + onsite exact exchange		
	PBE-SL0r	PBE; onsite SCAN-L/exact exchange		
	SCAN0r	SCAN + onsite exact exchange		
	r^{2}SCAN0r	r^{2}SCAN + onsite exact exchange		
for testing and reproducing old results				
−10		PBE before the 2024/2026 corrections (see text)		
−12		PBE exchange + PW91 correlation (old type 12)		
5002		PBE with exchange scaled by 0.75		
10001–7		single components (PZ, PW92, P86, PBE, PW91 correlation; PBE, RPBE exchange)		

Chapter 3

Plane waves and Brillouin-zone sampling

3.1 Introduction 10
3.2 Bloch states and plane waves 10
3.3 Cutoffs and FFT grids 10
3.4 k-point sampling 11
3.5 The Γ point 11

3.1 Introduction

Plane waves are the simplest basis set one can imagine. They are orthonormal, they do not depend on the positions of the atoms, and their completeness is controlled by a single number, the cutoff. The fast Fourier transform switches between real and reciprocal space at low cost, so that each term of the Hamiltonian can be applied where it is simplest. The price is that sharp features, such as the oscillations of the wave functions near a nucleus, would require an enormous number of plane waves. The PAW method of the next chapter removes this problem. Here we collect what the plane-wave description of a periodic solid involves: Bloch states, cutoffs, the grids for the density, and the sampling of the Brillouin zone.

3.2 Bloch states and plane waves

In a crystal with lattice vectors $\boldsymbol{T}$ the orbitals are Bloch states, $\psi_{n\boldsymbol{k}}(\boldsymbol{r}+\boldsymbol{T}) = e^{i\boldsymbol{k}\cdot\boldsymbol{T}}\psi_{n\boldsymbol{k}}(\boldsymbol{r})$, which are expanded in plane waves,

$$\tilde{\psi}_{n\boldsymbol{k}}(\boldsymbol{r}) = \sum_{\boldsymbol{G}} c_{n\boldsymbol{k}}(\boldsymbol{G})\, e^{i(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{r}}, \qquad \tfrac{1}{2}|\boldsymbol{k}+\boldsymbol{G}|^2 \le E_{\text{cut}}. \tag{3.1}$$

In PAW only the smooth pseudo wave functions $\tilde{\psi}$ are expanded in plane waves (Chapter 4).

3.3 Cutoffs and FFT grids

The wave-function cutoff is **`!CONTROL!FOURIER EPWPSI`** in Rydberg. The density contains products of wave functions and would need four times the cutoff; VESPAW uses **`EPWRHO = CDUAL`**×**`EPWPSI`**. **`CDUAL=4`** is exact, **`CDUAL=2`** is usually sufficient and cheaper. Density and

local potential live on a real-space grid; the Hamiltonian is applied to a wave function by transforming it to real space with a fast Fourier transform, multiplying by the potential and transforming back. In VESPAW these transforms are batched over blocks of 32 bands.

Energies converge from above with the cutoff, but energy *differences*, forces and stress converge faster. The required cutoff is set by the hardest PAW setup in the system; the setup generator prints an estimate (Sec. 4.5).

3.4 k-point sampling

Brillouin-zone integrals are replaced by sums over a Monkhorst–Pack mesh [30], **`!STRUCTURE!KPOINTS DIV=`**n_1 n_2 n_3, with an optional shift, or by a target density **`R`**. Symmetry and time reversal ($\psi_{-\boldsymbol{k}} = \psi_{\boldsymbol{k}}^*$) reduce the number of irreducible points. Metals need dense meshes together with a finite electronic temperature; insulators converge quickly.

3.5 The Γ point

At $\boldsymbol{k} = 0$ the wave functions can be chosen real. CP-PAW exploits this by packing two real bands into one complex array ("super wave functions"), which halves the number of FFTs. An alternative half-sphere storage (**`GAMMASUPER`**) is experimental and not yet correct.

Further reading

The original work on the subject of this chapter is found in Refs. [30, 31].

Chapter 4

The projector augmented-wave method

4.1 Introduction 12
4.2 The PAW transformation 12
4.3 Compensation charges, core and local potential 13
4.4 Setup construction 14
4.5 Quality checks 14
4.6 The setup library 15
4.7 Core-level spectra 15

4.1 Introduction

A wave function in a solid has two faces. Between the atoms, where the chemical bonds form, it is smooth. Close to a nucleus it oscillates rapidly, because it must be orthogonal to the core states. The classic electronic-structure methods have chosen one face. Pseudopotentials replace the atomic region by a smooth effective potential and lose the true wave function there. Augmented-wave methods keep it, but at the price of a basis set that depends on energy and on the atomic positions. The PAW method of Blöchl unites both views through a linear transformation between smooth auxiliary wave functions, which are expanded in plane waves, and the true wave functions. This chapter describes the transformation, the construction of the PAW setups, and the checks that make a setup trustworthy.

4.2 The PAW transformation

The PAW method [1] generalizes the ideas of the pseudopotential [32] and of the linear augmented-wave methods [33]. It maps the true all-electron (AE) valence orbitals ψ_n, which oscillate strongly near the nuclei, onto smooth pseudo (PS) orbitals $\tilde{\psi}_n$ by a linear transformation $|\psi_n\rangle = \mathcal{T}|\tilde{\psi}_n\rangle$. The transformation differs from unity only inside augmentation spheres around the atoms, $\mathcal{T} = 1 + \sum_R \mathcal{T}_R$. In each sphere we choose AE partial waves $\phi_{R,i}$ (solutions of the atomic Schrödinger equation at reference energies) and PS partial waves $\tilde{\phi}_{R,i}$, which agree with $\phi_{R,i}$ outside an augmentation radius r_c, such that $|\phi_{R,i}\rangle = \mathcal{T}|\tilde{\phi}_{R,i}\rangle$. Inside the sphere a PS orbital is expanded in PS partial waves, $|\tilde{\psi}_n\rangle = \sum_i |\tilde{\phi}_{R,i}\rangle c_{n,i}$. Because $\mathcal{T}$ is linear, the AE orbital has the same coefficients in the AE partial waves, $\mathcal{T}|\tilde{\psi}_n\rangle = \sum_i |\phi_{R,i}\rangle c_{n,i}$. The coefficients are obtained with projector functions $\tilde{p}_{R,i}$, localized within r_c and dual to the PS partial waves, $\langle \tilde{p}_{R,i}|\tilde{\phi}_{R,j}\rangle = \delta_{ij}$, as

$c_{n,i} = \langle \tilde{p}_{R,i} | \tilde{\psi}_n \rangle$. This gives

$$\mathcal{T} = 1 + \sum_{R,i} \big(|\phi_{R,i}\rangle - |\tilde{\phi}_{R,i}\rangle \big) \langle \tilde{p}_{R,i}|, \qquad |\psi_n\rangle = |\tilde{\psi}_n\rangle + \sum_{R,i} \big(|\phi_{R,i}\rangle - |\tilde{\phi}_{R,i}\rangle \big) \langle \tilde{p}_{R,i} | \tilde{\psi}_n \rangle. \tag{4.1}$$

In words: inside each sphere the smooth part $\sum_i |\tilde{\phi}_{R,i}\rangle\langle \tilde{p}_{R,i}|\tilde{\psi}_n\rangle$ is removed and replaced by its AE counterpart. The expansion is exact for complete sets of partial waves; in practice one or two partial waves per angular momentum suffice. The index i collects the angular momentum (ℓ, m) and a counter ν over partial waves of the same ℓ. The number of partial waves per ℓ is set by **`!STRUCTURE!SPECIES NPRO`**.

Frozen core. The core states are taken from the isolated atom and kept frozen, $|\psi^c_{R,\alpha}\rangle = |\phi^c_{R,\alpha}\rangle$, where α runs over the core states of atom R. They are not strictly localized within the sphere. Expectation values of a local operator A then split into a plane-wave part, one-center corrections and a core term,

$$\langle A \rangle = \sum_n f_n \langle \tilde{\psi}_n | \tilde{A} | \tilde{\psi}_n \rangle + \sum_R \sum_{ij} D_{R,ij} \big(\langle \phi_{R,j} | A | \phi_{R,i} \rangle - \langle \tilde{\phi}_{R,j} | \tilde{A} | \tilde{\phi}_{R,i} \rangle \big) + \sum_{R,\alpha} \langle \phi^c_{R,\alpha} | A | \phi^c_{R,\alpha} \rangle,$$
$$D_{R,ij} = \sum_n f_n \langle \tilde{p}_{R,i} | \tilde{\psi}_n \rangle \langle \tilde{\psi}_n | \tilde{p}_{R,j} \rangle, \tag{4.2}$$

with the one-center density matrix $D_{R,ij}$. For the density ($A = |\boldsymbol{r}\rangle\langle\boldsymbol{r}|$) this gives

$$n(\boldsymbol{r}) = \tilde{n}(\boldsymbol{r}) + \sum_R \big(n^1_R(\boldsymbol{r}) - \tilde{n}^1_R(\boldsymbol{r}) \big), \qquad n^1_R = \sum_{ij} D_{R,ij}\, \phi_{R,j} \phi_{R,i} + n^c_R, \tag{4.3}$$

where $\tilde{n} = \sum_n f_n |\tilde{\psi}_n|^2$ (plus the pseudized core density) is smooth and represented on the plane-wave grid, while the one-center densities n^1_R and $\tilde{n}^1_R$ are represented on radial grids times spherical harmonics. The total energy becomes $E = \tilde{E} + \sum_R (E^1_R - \tilde{E}^1_R)$: a plane-wave term plus, for each atom, the difference of an AE and a PS one-center energy evaluated on radial grids. The orbitals are orthonormal with respect to the overlap operator

$$\hat{O} = 1 + \sum_{R,ij} |\tilde{p}_{R,i}\rangle \Delta O_{R,ij} \langle \tilde{p}_{R,j}|, \qquad \Delta O_{R,ij} = \langle \phi_{R,i} | \phi_{R,j} \rangle - \langle \tilde{\phi}_{R,i} | \tilde{\phi}_{R,j} \rangle, \tag{4.4}$$

so the Kohn–Sham equations take the generalized form $\tilde{H}|\tilde{\psi}_n\rangle = \epsilon_n \hat{O} |\tilde{\psi}_n\rangle$ with

$$\tilde{H} = -\tfrac{1}{2}\nabla^2 + \tilde{v}(\boldsymbol{r}) + \sum_{R,ij} |\tilde{p}_{R,i}\rangle \Delta H_{R,ij} \langle \tilde{p}_{R,j}|. \tag{4.5}$$

4.3 Compensation charges, core and local potential

The electrostatics of the one-center densities is handled by compensation charges $\hat{n}_R$ (Gaussians of width set by **`RCSM/RCOV`**) that reproduce the multipole moments of $n^1_R - \tilde{n}^1_R$, so that the long-range electrostatics is carried by the smooth density $\tilde{n} + \hat{n}$ on the plane-wave grid. The one-center expansions are truncated at $\ell \le$ **`LRHOX`**. Core electrons are frozen in their atomic state. For the exchange-correlation energy, which is non-linear in the density, a pseudized core density (**`!CORE POW RC`**) is added to the smooth density. The local pseudopotential is constructed by pseudizing the AE potential inside a radius (**`!POT POW RC`**).

4.4 Setup construction

A PAW *setup* contains, for one element, the partial waves, projectors, core densities and local potential. VESPAW builds setups at run time from a short parameter block (**!SPECIES!AUGMENT**), or reads them from the library `vespaw_pot`. Three construction types exist:

- **HBS**: the scheme of Blöchl's later CP-PAW versions. The first partial wave of each ℓ is the lowest valence state; further partial waves are obtained from the nodeless ("$\dot{\phi}$-like") construction at higher energies, bounded by a box of radius **RBOX/RCOV**.
- **NDLSS**: an older nodeless scheme.
- **KERKER**: Kerker-type pseudization.

The radii are given in units of the covalent radius of the element: **RCL/RCOV** for the partial waves, **RCSM/RCOV** for the compensation charge, **RC/RCOV** in **!POT** and **!CORE**.

Semicore states. When shallow core states are treated as valence (e.g. Cu 3*s*3*p*, **ZV=19**), an angular momentum has two valence states, 3*s* and 4*s*. Each valence bound state then receives its own partial wave at its own energy; only further partial waves are box states. With a semicore state below it, the PS 4*s* orbital need not have a node, because it is orthogonal to the PS 3*s* only through $\hat{O}$. The setup test therefore identifies PS states by their energy, not by their node count.

4.5 Quality checks

A setup is an approximation, and a bad one fails silently: calculations converge to wrong answers. The generator therefore reports three checks.

Ghost test. The atomic PAW Hamiltonian of each ℓ is diagonalized in a basis of 120 spherical Bessel functions inside a large sphere. Its lowest levels must match the AE valence levels. A level far below the lowest AE reference is a *ghost state*; the generator prints `WARNING: GHOST STATE IN SETUP`. The overlap matrix in this basis must be positive definite; if it is not, the setup is unusable.

Incompleteness warning. With only one partial wave per ℓ, the local potential may bind a state far below the valence level whose wave function lies largely outside the span of the projectors. Such components receive no one-center correction. In a solid they can sink into the potential well of the compensation charge, and the calculation converges to a collapsed state far below the physical minimum. This happened for NiO with a single-projector Ni setup: the collapse deepened with the cutoff and disappeared with a second *s* projector. The generator prints `WARNING: INCOMPLETE SETUP` when the lowest state of the local potential lies more than 0.5 H below the valence reference with a single projector.

Kinetic-energy cutoff table. As in the VASP setup generator, the kinetic energy of each PS valence orbital that lies above a cutoff is computed from its Fourier transform, weighted by the occupation and summed. The generator prints the error as a function of the cutoff, from 20 eV to about 290 Ry, and suggests `ENMAX` and `ENMIN` (errors of 6 and 13 meV), or states that they are not reached. The PS orbitals are tapered to zero before the box wall to avoid a spurious kink.

4.6 The setup library

`vespaw_pot` contains one species block per element and variant. Its audit (`vespaw_pot/AUDIT.md`) lists each setup with its status (OK, incomplete, ghost, unusable) and the recommended replacement. The flagged setups were rebuilt as HBS setups with two projectors per ℓ. Validation in solids shows that the frozen-semicore $3d$ setups (Fe, Cu) reproduce AE-PBE lattice constants only to about 1%, and that the error grows with the augmentation radius; a semicore Cu setup with small radii reaches 0.5% at a high cutoff. Accurate soft $3d$ setups are an open problem (see the audit).

4.7 Core-level spectra

Because PAW keeps the AE partial waves and the frozen core states, matrix elements between core and band states are available. VESPAW uses them for electron-energy-loss near-edge spectra (ELNES), with **`!CONTROL!ANALYSE!EELS`**. The same analysis applies to x-ray absorption near-edge spectra (XANES), which also probe the excitation of a core electron into empty states. The spectrum is the loss function

$$L(\omega) = -\mathrm{Im}\left[\epsilon_r^{-1}(\omega)\right], \qquad \epsilon_r(\omega) = 1 + 4\pi\frac{N}{V}\sum_n (1-f_n)\frac{Q_n}{(\epsilon_n-\epsilon_c)^2-\omega^2-i\Gamma\omega}, \tag{4.6}$$

following Wooten [34], where N/V is the number density of the probed atom, ϵ_c the core level and ϵ_n the energy of an empty Kohn–Sham state. The oscillator strengths are

$$Q_n = 2\sum_c (\epsilon_n-\epsilon_c)\,\frac{1}{3}\sum_{j=1}^{3}\left|\langle\psi_c|x_j|\psi_n\rangle\right|^2, \tag{4.7}$$

where c runs over the states of the core shell (**`IC`**). The core state is localized, so the band state can be replaced by its one-center expansion, $|\psi_n\rangle \approx \sum_i |\phi_{R,i}\rangle\langle\tilde p_{R,i}|\tilde\psi_n\rangle$. With $\phi_{R,i}(\boldsymbol{r}) = R_i(r)Y_{L_i}(\hat r)$ and the dipole operator written with real spherical harmonics, the matrix elements reduce to radial integrals and Gaunt coefficients $C_{L,L',L''} = \int d\Omega\, Y_L Y_{L'} Y_{L''}$,

$$\langle\psi_c|x_j|\psi_n\rangle = \sqrt{\frac{4\pi}{3}}\sum_i C_{L_c,p_j,L_i}\int dr\, r^3 R_c(r)R_i(r)\,\langle\tilde p_{R,i}|\tilde\psi_n\rangle, \tag{4.8}$$

where p_j denotes the $\ell = 1$ harmonic along x_j. The spectrum therefore measures the empty density of states on the probed atom with angular momentum $\ell_c \pm 1$.

Broadening. The width Γ is the sum of the core-hole and the band-state lifetime broadenings. The core-hole width of a K shell is taken from Eq. 12 of Egerton [35], $\Gamma_K = [-0.285 + 0.0216\,(E_{\mathrm{thr}}/\mathrm{eV})^{0.472}]$ eV, with the threshold energy $E_{\mathrm{thr}} = -\epsilon_c$; other shells use Eq. 14 of the same reference. The band state with energy ϵ above the Fermi level decays with the rate v/λ, where $v = \sqrt{2\epsilon}$ is the free-electron velocity and λ the inelastic mean free path, interpolated as $\lambda = A/\epsilon^2 + B\sqrt{\epsilon}$ with $A = 641$ eV^2 nm and $B = 0.096$ nm eV$^{-1/2}$ for inorganic compounds [36]. The result is finally convoluted with a Gaussian for the instrumental resolution (**`BROADENING[EV]`**).

Limitations. The calculation uses the independent-particle approximation and ignores the core hole; its only effect is a rigid shift of the spectrum, which must be adjusted to the experimental threshold. A better approximation is a supercell calculation with half a core hole on the probed atom (Slater's transition-state rule), and a still better one is the solution of the Bethe–Salpeter

equation [37]. The independent-particle approximation is adequate for the oxygen K edge ($1s \rightarrow 2p$), where the core hole is strongly screened [38]. It is not adequate for transition-metal $L_{2,3}$ edges, whose shape is dominated by multiplet effects from the overlap of the $2p$ hole with the $3d$ electrons.

Further reading

The original work on the subject of this chapter is found in Refs. [1, 35, 39].

Chapter 5

Local hybrid functionals

5.1 Introduction . . . 17
5.2 Hybrid functionals . . . 17
5.3 Local orbitals . . . 18
5.4 Projection onto the local orbitals . . . 18
5.5 PBE0r . . . 18
5.6 PBE-SL0r . . . 19
5.7 SCAN0r . . . 19

5.1 Introduction

Hybrid functionals admix a fraction a_x of exact exchange and cure much of the band-gap problem of semilocal functionals, but in a plane-wave basis exact exchange is one to two orders of magnitude more expensive than a GGA. Local hybrids of the PBE0r type [40, 41] keep only the onsite part of the exchange correction, evaluated in a basis of local orbitals, at nearly semilocal cost.

5.2 Hybrid functionals

The adiabatic connection, Eq. (2.13), writes the exchange-correlation energy as an integral of $U_{\mathrm{xc}}^{\lambda}$ from the non-interacting ($\lambda = 0$) to the fully interacting ($\lambda = 1$) system. At $\lambda = 0$ the integrand is the exact exchange energy of the Kohn–Sham determinant; near $\lambda = 1$ a semilocal functional is a good approximation. Becke [42] approximated the integral by the trapezoidal rule, $E_{\mathrm{xc}} \approx \frac{1}{2}E_x^{\mathrm{HF}} + \frac{1}{2}U_{\mathrm{xc}}^{\lambda=1}$, and obtained the "half-and-half" functional. If the integrand is a convex rather than a straight function of λ, a smaller weight of exact exchange is appropriate. The general form of a hybrid is

$$E_{\mathrm{xc}}^{\mathrm{hybrid}} = E_{\mathrm{xc}}^{\mathrm{GGA}} + a_x\left(E_x^{\mathrm{HF}} - E_x^{\mathrm{GGA}}\right). \tag{5.1}$$

Perdew, Ernzerhof and Burke [43] argued from perturbation theory that $a_x = \frac{1}{4}$ is a good choice with PBE, which gives PBE0 [44]. The most widely used hybrid in chemistry, B3LYP [45], is a generalization of Becke's three-parameter form [46]. Hybrids improve binding energies, reaction energies and band gaps over GGAs. In transition-metal oxides they open the gaps that semilocal functionals close, because exact exchange stabilizes occupied orbitals relative to empty orbitals of the same character, and removes the self-interaction of the localized d and f electrons.

5.3 Local orbitals

Each atom carries natural tight-binding orbitals (NTBOs) χ_a [40, 47]. Inside the augmentation sphere of radius **RAUG/RCOV** the head function of an orbital is an augmented screened Hankel function, a combination of an AE partial wave ϕ and its energy derivative $\dot\phi$,

$$K^{\mathrm{aug}} = \phi\, k_\phi + \dot\phi\, k_{\dot\phi}, \tag{5.2}$$

matched to a Hankel function with κ^2 set by **K2**. In the spheres of the neighbors it is continued by screened Bessel functions $\bar J^{\mathrm{aug}} = \dot\phi\, j_{\dot\phi}$ with the structure constants $\bar S$. Exponential tails (**TAILLAMBDA**, **RTAIL/RCOV**) and a cutoff (**RTAILCUT/RCOV**) make the orbitals short ranged. The orbitals considered per ℓ are selected by **NOFL**.

The construction follows the tight-binding linear muffin-tin orbital method [33, 47]: only the nodeless scattering partial waves are admitted on the neighboring sites, the same framework of nodeless partial waves that is used for the augmentation [48].

5.4 Projection onto the local orbitals

The onsite density matrix $D^\sigma_{ab} = \sum_n f_{n\sigma}\langle\chi_a|\psi_{n\sigma}\rangle\langle\psi_{n\sigma}|\chi_b\rangle$ requires the coefficients of the wave functions in the local-orbital basis. Inside the spheres the one-center expansion $\psi^1 = \sum_i \phi_i\langle\tilde p_i|\tilde\psi\rangle$ is fitted by local orbitals in the least-squares sense,

$$\boldsymbol{c} = \big[\langle\chi|\Theta|\chi\rangle\big]^{-1}\langle\chi|\Theta|\phi\rangle\langle\tilde p|\tilde\psi\rangle, \tag{5.3}$$

where Θ restricts integrals to the spheres. Both overlaps are computed with the exact χ inside the spheres, i.e. with K^{aug} and $\bar J^{\mathrm{aug}}$ as radial functions.

Pitfall. The original implementation represented χ by its partial-wave expansion $\langle\tilde p|\tilde\chi\rangle$. The $\dot\phi$ component of χ converges slowly in that expansion, so the fitted orbital lacked norm and the density matrix overcounted charge: 13–15 electrons on Ni in NiO instead of about 7. The hybrid energy and the gap then changed with the number of projectors (NiO PBE0r gap 3.96, 4.08, 4.10 eV for 2, 3, 4 d projectors). With the exact overlaps they are converged (2.99, 3.00, 3.00 eV). The old fit can be restored with the environment variable `VESPAW_NTBO_OLDPROJ=1` for comparison.

5.5 PBE0r

PBE0r keeps only the onsite terms of the PBE0 exchange correction,

$$E_{\mathrm{xc}}^{PBE0r} = E_{\mathrm{xc}}^{\mathrm{PBE}}[n] + a_x\big(E_x^{\mathrm{HF,1c}} - E_x^{\mathrm{PBE,1c}}\big), \qquad E_x^{\mathrm{HF,1c}} = -\frac{1}{2}\sum_\sigma\sum_{abcd} D^\sigma_{ac} D^\sigma_{bd} U_{abcd}, \tag{5.4}$$

with the onsite Coulomb tensor U of the local orbitals. The double-counting term $E_x^{\mathrm{PBE,1c}}$ is the PBE exchange of the local-orbital density, obtained from the full one-center density with the weight $w(r) = (n_{\mathrm{loc}}/n_{\mathrm{all}})^2$. Core–valence exchange can be included (**CV=T**). The mixing a_x is set globally by **!CONTROL!DFT!NTBO HFWEIGHT** and per species by **LHFWEIGHT**. The exchange correction enters the Hamiltonian in two places: the one-center PAW Hamiltonian ΔH (derivative with respect to the full one-center density) and an operator in the local-orbital basis (derivative with respect to D).

Design choices. In contrast to other local hybrids [49, 50], which correct only the d or f shell of selected atoms, PBE0r includes the correction on all atoms and for all local orbitals that contribute to the valence or contain core states, including the exchange between core and valence electrons. Applying the correction to a subset of orbitals would shift them in energy relative to the uncorrected ones; including all orbitals avoids such artificial shifts. Because the correction is onsite, the mixing factor can be chosen per atom, which reflects an effective local dielectric screening. The common value $a_x = 0.25$ is not always appropriate: in Pr manganites it made the observed configuration of the Pr $4f$ shell unstable, and the mixing was adjusted per species to reproduce the observed spectral features [41].

5.6 PBE-SL0r

PBE-SL0r (called SCAN0r before October 2026) replaces the semilocal onsite exchange by SCAN-L exchange while keeping PBE for the delocalized part and for all correlation:

$$\begin{aligned} E_{\mathrm{xc}}^{PBE-SL0r} &= E_{\mathrm{xc}}^{\mathrm{PBE}}[n] - E_x^{\mathrm{PBE,1c}} + a_x E_x^{\mathrm{HF,1c}} + (1-a_x) E_x^{\mathrm{SCAN,1c}} \\ &= E_{\mathrm{xc}}^{\mathrm{PBE}}[n] + \underbrace{\left(E_x^{\mathrm{SCAN,1c}} - E_x^{\mathrm{PBE,1c}}\right)}_{\text{swap term}} + a_x\left(E_x^{\mathrm{HF,1c}} - E_x^{\mathrm{SCAN,1c}}\right). \end{aligned} \tag{5.5}$$

The swap term is not scaled by a_x. PBE-SL0r is selected by **`!NTBO MODUS='HYBRID' HFWEIGHT=`**a_x **`DCTYPE='SCANL'`** with a PBE base functional. A consistent implementation requires:

1. the swap term in both Hamiltonian channels (one-center and local-orbital), at full weight;
2. the full Euler–Lagrange potential of the onsite SCAN-L exchange, Eq. (2.18), with the weight w inside the derivatives;
3. exact spin scaling, Eq. (2.19). Evaluating the onsite SCAN-L exchange for the spin-summed density replaces spin-polarized PBE exchange by unpolarized SCAN-L exchange in the swap term and removes a fraction $1 - a_x$ of the onsite exchange splitting; magnetic gaps collapse (NiO 1.0 instead of 2.0 eV);
4. the exact local-orbital projection of Sec. 5.4.

All four were found by checking that the variational optimizer and damped dynamics reach the same state (Sec. 6.7).

5.7 SCAN0r

SCAN0r uses SCAN with the orbital kinetic-energy density everywhere, for the plane-wave density and in the spheres, for exchange and correlation, and replaces a fraction a_x of the onsite SCAN exchange of the local orbitals by exact exchange:

$$E_{\mathrm{xc}}^{SCAN0r} = E_{\mathrm{xc}}^{\mathrm{SCAN}}[n,\tau] + a_x\left(E_x^{\mathrm{HF,1c}} - E_x^{\mathrm{SCAN,1c}}\right). \tag{5.6}$$

It is the meta-GGA analogue of PBE0r, Eq. (5.4), without a swap term, and is selected by **`TYPE='SCAN' TSCANDELOC=T`** with **`!NTBO MODUS='HYBRID' HFWEIGHT=`**a_x. The double counting $E_x^{\mathrm{SCAN,1c}}$ uses the onsite τ of the local orbitals, is spin scaled, and by default contains exchange only (**`TSCANCORR=T`** adds SCAN correlation). Exchange-only spin scaling and the spin-polarized correlation of Chapter 2 apply as for SCAN. With **`TYPE='R2SCAN'`** the same construction gives r^{2}SCAN0r, with r^{2}SCAN everywhere and in the double counting.

Further reading

The original work on the subject of this chapter is found in Refs. [40–44].

Chapter 6

Self-consistency: the variational optimizer

6.1 Introduction 21
6.2 The problem with density mixing 21
6.3 Free energy as a function of the Hamiltonian 22
6.4 The gauge direction 22
6.5 Line search and acceptance 22
6.6 Route 2: direct joint minimization 23
6.7 The consistency test 23

6.1 Introduction

The Kohn–Sham equations are a fixed-point problem: the density determines the potential, and the potential determines the density. In almost every plane-wave code this problem is solved by mixing input and output densities until they agree, a procedure that works well in most cases and fails, when it fails, without warning. The theory, however, offers a minimum principle. This chapter shows how the self-consistency problem can be written as a minimization in which the effective Hamiltonian is the variable, and every energy is that of an actual state. It describes the structure of this Hamiltonian space, including a gauge mode that is specific to the PAW method, and the line search that turns it into a robust algorithm.

6.2 The problem with density mixing

The standard SCF cycle alternates between solving the Kohn–Sham equations in a fixed potential and building a new input potential by mixing the input and output densities or potentials (Pulay/DIIS [51], Anderson [52], with Kerker preconditioning [53]). The energy along this path is a Harris–Foulkes-type estimate [54, 55]; it is not an upper bound and need not decrease. Mixing can stagnate, oscillate, or converge to an excited state, and nothing in the iteration signals it. VESPAW offers density mixing (**`MIXER='PULAY'`** or **`'ANDERSON'`**) but uses a variational optimizer by default.

6.3 Free energy as a function of the Hamiltonian

We minimize the Mermin free energy, Eq. (2.20), in the PAW form. The parameters of the Hamiltonian are collected in a vector

$$\boldsymbol{\theta} = \big(\tilde{v}(\boldsymbol{r}),\ v_\tau(\boldsymbol{r}),\ \{\Delta H_{R,ij}\}\big). \tag{6.1}$$

For given $\boldsymbol{\theta}$ we determine the eigenstates $\psi_n(\boldsymbol{\theta})$ of $\tilde{H}[\boldsymbol{\theta}]$ and the Fermi occupations $f_n(\boldsymbol{\theta})$ at fixed N, and define

$$\mathcal{A}(\boldsymbol{\theta}) = A\big[\{\psi_n(\boldsymbol{\theta})\}, \{f_n(\boldsymbol{\theta})\}\big]. \tag{6.2}$$

Because $\psi_n(\boldsymbol{\theta})$ and $f_n(\boldsymbol{\theta})$ form an admissible ensemble, $\mathcal{A}(\boldsymbol{\theta})$ is an upper bound of the minimum of A for every $\boldsymbol{\theta}$. At self-consistency, where $\boldsymbol{\theta}$ equals the parameters $\boldsymbol{v}_{\text{out}}[\boldsymbol{\theta}]$ built from the output state, the bound is attained. Following Gonze [56],

$$\frac{\partial \mathcal{A}}{\partial \boldsymbol{\theta}} = \chi(\boldsymbol{\theta})\,\big(\boldsymbol{v}_{\text{out}}[\boldsymbol{\theta}] - \boldsymbol{\theta}\big), \tag{6.3}$$

with χ the static independent-particle response. Since χ is negative semidefinite, the residual $\boldsymbol{r} = \boldsymbol{v}_{\text{out}} - \boldsymbol{\theta}$, and any positive-definite preconditioned version $K\boldsymbol{r}$, is a descent direction; χ never has to be built. The Kerker filter $K(G) = G^2/(G^2 + q_0^2)$ is applied to the plane-wave part of the residual, with q_0^2 set by **MIXGOSQ** (default 1.5 bohr^{-2}). The default is chosen for solids. For molecules and slabs with large vacuum it damps the long-wavelength components so strongly that the gains of a step fall below the noise of the iterative eigensolver, and the line search stalls; use $q_0^2 \approx 0.05$–0.1 there.

6.4 The gauge direction

A uniform shift $\tilde{H} \to \tilde{H} + c\hat{O}$ shifts all eigenvalues and the Fermi level by c and leaves orbitals and occupations unchanged. In parameter space it is the direction $\boldsymbol{s} = (1, 0, \{\Delta O_{R,ij}\})$, and $\mathcal{A}(\boldsymbol{\theta} + c\boldsymbol{s}) = \mathcal{A}(\boldsymbol{\theta})$ exactly. The two halves of $\boldsymbol{s}$ are updated by different rules (Kerker on the grid, damped steps in ΔH). If $\boldsymbol{s}$ is not treated explicitly, they drift apart and act as a spurious potential step between spheres and interstitial (Si: convergence to an excited state 28 mH above the ground state). VESPAW therefore

1. projects $\boldsymbol{s}$ out of every residual, and
2. after each step moves $\boldsymbol{\theta}$ along $\boldsymbol{s}$ so that its mean grid potential equals that of the output state.

The second step does not change any energy, but without it $\boldsymbol{\theta}$ keeps its starting gauge while the true potential moves: eigenvalues and Fermi level are then offset (0.49 eV in Si), and with them the Lagrange multipliers in forces and stress (Chapter 8).

6.5 Line search and acceptance

Steps are proposed by an Anderson extrapolation built from *accepted* points only. A trial $\boldsymbol{\theta}'$ is accepted if $\mathcal{A}(\boldsymbol{\theta}') \le \mathcal{A}_{\text{best}} + \epsilon$; otherwise the step is halved towards the best point. After five consecutive rejections the best point is re-evaluated ("re-anchoring"), because its energy may stem from an eigensolution that was not fully converged.

Convergence is reached when the mean-square grid residual is below **MIXTOL** (and the one-center residual below its square root), or when two consecutive accepted energies differ by less than **EDIFF**. Route 1 is selected by **!PSIDYN MIXER='VARIATIONAL'** together with **OPTIMIZER='DAV'**, **'RMM'** or **'CG'**; it is the default mixer.

Algorithm 1 Hamiltonian-space variational SCF (Route 1).

$\boldsymbol{\theta}_0 \leftarrow \boldsymbol{v}_{\text{out}}[\psi_{\text{start}}]$; hold fixed for a short eigensolver warm-up
loop
  diagonalize $\tilde{H}[\boldsymbol{\theta}]$ (Davidson, RMM-DIIS or CG); Fermi occupations at fixed N
  evaluate $\mathcal{A}(\boldsymbol{\theta})$ at the output state; build $\boldsymbol{v}_{\text{out}}$
  $\boldsymbol{r} \leftarrow \boldsymbol{v}_{\text{out}} - \boldsymbol{\theta}$; project out $\boldsymbol{s}$
  if $\mathcal{A}(\boldsymbol{\theta}) \leq \mathcal{A}_{\text{best}} + \epsilon$ **then**
    accept; add $(\boldsymbol{\theta}, K\boldsymbol{r})$ to the history; stop if converged
    $\boldsymbol{\theta} \leftarrow$ Anderson proposal
  else
    $\boldsymbol{\theta} \leftarrow \frac{1}{2}(\boldsymbol{\theta} + \boldsymbol{\theta}_{\text{best}})$ ▷ re-anchor after 5 rejections
  end if
  re-gauge $\boldsymbol{\theta}$ along $\boldsymbol{s}$ to the mean grid potential of $\boldsymbol{v}_{\text{out}}$
end loop

Hybrid operators. The onsite hybrid operator of Chapter 5 depends on the orbitals through D, not only on the density. It is taken from the current state rather than line-searched; including its matrix elements in $\boldsymbol{\theta}$ left a residual that did not decay.

6.6 Route 2: direct joint minimization

As an alternative (**`OPTIMIZER='EDFT'`**), A is minimized directly with respect to the orbitals and an auxiliary Hamiltonian H^{aux} whose eigenvalues define the occupations, in the spirit of ensemble DFT [57, 58]. The orbital direction is the Teter-preconditioned [59] projected residual combined by Polak–Ribière conjugate gradients; the direction for H^{aux} is $H^{\text{phys}} - H^{\text{aux}}$. Trial points are built by Löwdin orthonormalization and rotated into the eigenbasis of H^{aux}; step lengths come from a quadratic model, with the same acceptance and re-anchoring logic as Route 1. The orbital direction is computed from a Kerker-preconditioned Hamiltonian. With this preconditioning Route 2 currently stops about 5×10^{-6} H above the minimum on Si; without it (`VESPAW_EDFTVAR_NOKERKER=1`) it reaches the minimum.

6.7 The consistency test

The variational optimizer minimizes the energy. Damped Car–Parrinello dynamics (Chapter 7) and density mixing converge to a self-consistent state. The two coincide only if the Hamiltonian is the derivative of the energy. Agreement between them, to 10^{-6} H or better, is therefore a sharp test of any new functional or implementation: a missing potential term shows up as the variational optimizer finding energies *below* the self-consistent state without converging. The regression suite (Chapter 9) applies this test to every case.

Further reading

The original work on the subject of this chapter is found in Refs. [56–58, 60].

Chapter 7

Eigensolvers and dynamics

7.1 Introduction 24
7.2 Block Davidson 24
7.3 RMM-DIIS 25
7.4 Conjugate gradients 25
7.5 Car–Parrinello dynamics with friction 25

7.1 Introduction

Most of the computing time of a calculation is spent in finding the lowest eigenstates of the Hamiltonian. For a few hundred bands in a basis of tens of thousands of plane waves, a direct diagonalization is out of the question. Iterative methods instead improve a set of trial wave functions, using only the action of the Hamiltonian on them. In the original Car–Parrinello approach this is done by a fictitious dynamics of the wave functions. This chapter describes the iterative eigensolvers of VESPAW and the damped dynamics of CP-PAW, and what each of them requires to return reliable orthonormal eigenstates.

For Route 1 the eigenstates of $\tilde{H}[\boldsymbol{\theta}]$ must be found at every iteration. They are warm-started from the previous iteration and converged iteratively. All solvers work with the generalized problem $\tilde{H}\psi = \epsilon\hat{O}\psi$ and apply $\tilde{H}$ and $\hat{O}$ in blocks of bands.

7.2 Block Davidson

`OPTIMIZER='DAV'`. Each band is extended by a preconditioned correction vector $(\tilde{H} - \epsilon\hat{O})\psi$; the new vectors are projected out of *all* current bands (twice), vectors lost to cancellation are dropped, and the extended basis is canonically orthogonalized with an eigenvalue cutoff of 10^{-10}. A Rayleigh–Ritz step in this subspace gives the new bands. Converged bands are locked. The implementation keeps seven wave-function-sized arrays. Davidson is robust from any start and is the reference solver.

Pitfall. An earlier Davidson orthogonalized only against converged bands and stabilized the near-singular overlap with a shift; the returned orbitals were not orthonormal, which produced energies below the minimum and split degenerate states.

7.3 RMM-DIIS

OPTIMIZER='RMM' . Residual minimization with direct inversion in the iterative subspace [61, 62] minimizes the norm of the residual of each band separately, without orthogonalization against other bands, and is therefore cheap and parallel. It converges to the eigenvector nearest the starting guess. Started from random orbitals it converges to wrong states, so the first **RMMDELAY** iterations (default 8) use Davidson. Bands are processed in blocks of 32 (**NRMMX** inner sweeps). RMM-DIIS is the fastest option for large systems (Si_{64}: 247 s vs 386 s for Davidson).

Pitfall. With the onsite hybrid, RMM-DIIS reaches the correct energy but can miss the lowest empty band (MgO PBE-SL0r gap 6.54 instead of 4.23 eV). Take band gaps of hybrid calculations from Davidson.

7.4 Conjugate gradients

OPTIMIZER='CG' . Band-by-band preconditioned conjugate-gradient minimization of the Rayleigh quotient with orthogonalization to the lower bands, followed by a subspace rotation. It is reliable, but slower than the block methods for many bands.

7.5 Car–Parrinello dynamics with friction

The original CP-PAW optimizer (**OPTIMIZER='MD'** , the default when no eigensolver is chosen) treats the wave-function coefficients as classical degrees of freedom with a fictitious mass m_ψ [63]. Instead of solving the electronic problem at each atomic configuration, as in Born–Oppenheimer dynamics, Car and Parrinello propagate orbitals and nuclei together with the Lagrangian

$$\mathcal{L} = \sum_n f_n\, m_\psi \langle \dot{\tilde\psi}_n | \dot{\tilde\psi}_n \rangle + \sum_I \tfrac{1}{2} M_I \dot{\boldsymbol{R}}_I^2 - E\big[\{\tilde\psi_n\}, \{\boldsymbol{R}_I\}\big] + \sum_{nm} \Lambda_{mn} \big(\langle \tilde\psi_n | \hat{O} | \tilde\psi_m \rangle - \delta_{nm} \big). \tag{7.1}$$

The terms are the fictitious kinetic energy of the orbitals, the kinetic energy of the nuclei, the total energy, and the orthonormality constraint with Lagrange multipliers Λ. The Euler–Lagrange equations give Newton's equations for the nuclei, with the forces of Chapter 8, and, after division by f_n and absorbing the occupations into Λ, the equation of motion of the orbitals with an added friction term,

$$m_\psi \ddot{\tilde\psi}_n = -\tilde{H}\tilde\psi_n + \sum_m \hat{O}\tilde\psi_m \Lambda_{mn} - \gamma\, m_\psi \dot{\tilde\psi}_n, \tag{7.2}$$

with a friction γ (**!PSIDYN FRIC**). With friction the dynamics relaxes into the ground state. The "autopilot" (**!PSIDYN!AUTO**) lowers the friction when the energy decreases and raises it when it increases. Damped dynamics is slow but very robust, converges to the self-consistent state for any functional, and serves as the reference in the consistency test. **SAFEORTHO=T** orthogonalizes such that the energy is conserved; it should not be used with variable occupations.

Further reading

The original work on the subject of this chapter is found in Refs. [31, 61–64].

Chapter 8

Forces, stress and molecular dynamics

8.1 Introduction 26
8.2 Forces 26
8.3 Stress 27
8.4 Structure relaxation and molecular dynamics 27

8.1 Introduction

Once the electrons are in their ground state, the atoms feel forces and the cell feels a stress. These derivatives of the energy are what makes structure optimization and molecular dynamics possible. In a variational theory they follow from the explicit dependence of the energy on the atomic positions, but in the PAW method this dependence hides in many places: the projector functions, the compensation charges and the overlap operator all move with the atoms. This chapter collects the expressions for forces and stress and describes the equations of motion for the atoms.

8.2 Forces

The force on atom R is the total derivative of the energy with respect to its position. At a stationary point of the energy the orbital response drops out (Hellmann–Feynman [4]). In PAW the projectors, the compensation charges and the one-center terms move with the atom, and the orthonormality constraint depends on the positions through $\hat{O}$:

$$\boldsymbol{F}_R = -\frac{\partial E}{\partial \boldsymbol{R}}\Big|_{\text{explicit}} - \sum_{nm} \Lambda_{mn} \, \langle \tilde{\psi}_n | \frac{\partial \hat{O}}{\partial \boldsymbol{R}} | \tilde{\psi}_m \rangle \times (\text{occupation weights}), \tag{8.1}$$

where the first term contains the derivatives of ΔH, the projectors and the compensation charges. In Car–Parrinello dynamics Λ comes from the orthonormality constraint. For the eigensolvers the orbitals are Ritz vectors, and $\Lambda = \mathrm{diag}(\epsilon_n)$.

Pitfalls. Before September 2026 the eigensolvers did not set Λ at all, so the constraint term was missing (0.5% error in the Si force), and with the variational mixer the eigenvalues carried the gauge offset of the previous chapter. Both are now fixed, and forces agree with finite differences of the energy to 10^{-4} relative (regression case `si_forces`).

8.3 Stress

The stress tensor is the derivative of the energy with respect to a homogeneous strain ϵ,

$$\sigma_{\alpha\beta} = -\frac{1}{V}\frac{\partial E}{\partial \epsilon_{\alpha\beta}}, \tag{8.2}$$

following Nielsen and Martin [65]; positive values mean that the cell wants to expand, and the pressure is $\mathrm{tr}\,\sigma/3$. It contains kinetic, Hartree, exchange-correlation, projector and compensation-charge terms, and the same constraint term as the forces.

The analytic stress is taken at a fixed set of plane waves (the basis strains with the cell). A finite difference at fixed cutoff changes the basis and contains the *Pulay stress*, which vanishes as the cutoff grows (Si: 2% at 25 Ry, 0.5% at 50 Ry). Lattice constants from energy-volume curves need converged cutoffs for the same reason.

Output. **`!CONTROL!GENERIC FORCES=T STRESS=T`** evaluates forces and stress also for fixed atoms and a fixed cell and prints them at full precision at the end of the run (`FORCE`, `STRESS`, `PRESSURE[GPA]` lines).

8.4 Structure relaxation and molecular dynamics

Atomic motion is controlled by **`!CONTROL!RDYN`**. With friction (**`FRIC`**, **`!RDYN!AUTO`**) the atoms relax into a minimum; without friction they follow Newton's equations. The cell shape can be relaxed or propagated with **`!CONTROL!CELL MOVE=T`** [65].

Car–Parrinello vs Born–Oppenheimer MD. In Car–Parrinello MD the wave functions are propagated together with the atoms and stay close to the ground state if their fictitious mass is small enough (adiabaticity [66]). The time step must resolve the fastest wave-function motion. In Born–Oppenheimer MD the electrons are converged at each step (with an eigensolver and Route 1); the time step is set by the ions, but each step costs a full SCF.

Thermostats. Temperature is controlled by Nosé–Hoover thermostats [67] on the atoms (**`!RDYN!THERMOSTAT[K]`**) and, in Car–Parrinello MD, on the wave functions [68] (**`!PSIDYN!THERMOSTAT`**), which removes the heat that leaks from the ions into the fictitious electronic motion.

Further reading

The original work on the subject of this chapter is found in Refs. [65–68].

Chapter 9

Structure of the code

9.1 Introduction 28
9.2 Source files 28
9.3 One SCF step 29
9.4 Input 29
9.5 Adding a functional 29
9.6 Adding an optimizer 29
9.7 Regression suite 29

9.1 Introduction

This chapter is written for those who want to read the source code, change it, or add to it. It gives an overview of the source files, follows one self-consistency step through the program, and describes how a new functional or a new optimizer is added and tested.

9.2 Source files

File	Content
`vespaw.f90`, `vespaw_driver.f90`	main program, time-step loop, reports
`vespaw_ioroutines.f90`	reading `.cntl/.strc` input
`vespaw_waves1.f90`, `vespaw_waves2.f90`	wave functions, density, $H\psi$, forces, total energy
`vespaw_cg.f90`	Davidson, RMM-DIIS, CG
`vespaw_mixer.f90`	variational, Pulay and Anderson mixers
`vespaw_edft.f90`	Route 2 (EDFT)
`vespaw_potential.f90`	plane-wave potential, electrostatics, XC on the grid
`vespaw_augmentation.f90`	one-center (PAW sphere) terms
`vespaw_setups.f90`	setup construction and tests
`vespaw_lmto.f90`	local orbitals, PBE0r, PBE-SL0r, SCAN0r
`vespaw_scantau.f90`, `vespaw_scannsph.f90`	SCAN with orbital τ, non-spherical one-center SCAN
`vespaw_dft.f90`	exchange-correlation functionals
`vespaw_atoms.f90`, `vespaw_cell.f90`	ionic and cell dynamics
`vespaw_occupations.f90`	occupations, Mermin

The code is organized in CP-PAW's object style: each module keeps its data in a Fortran module and is accessed through `OBJECT$SET`, `$GET` and action routines.

9.3 One SCF step

1. `PAW` (driver) calls `TIMESTEP`, which calls `WAVES$ETOT`.
2. `WAVES$ETOT`: projections $\langle\tilde{p}|\tilde{\psi}\rangle$, density and one-center density matrices; plane-wave potential (`WAVES_VOFRHO`); one-center potentials and ΔH (`AUGMENTATION`); hybrid terms (`LMTO`); forces and stress (`WAVES$FORCE`).
3. Variational mixer (`WAVES_VARMIXHAMILTONIAN`): the output Hamiltonian and the energy are handed to the line search, which returns $\boldsymbol{\theta}$ for the next diagonalization.
4. Eigensolver (`DAVIDSON$STATE_BY_BLOCK`, `RMM$STATE_BY_BLOCK`, `CG$STATE_BY_STATE`) or, for dynamics, $H\psi$ and propagation.
5. Reports (`PRINFO`), capture of forces and stress, and the switch of all dynamical variables to the next step.

9.4 Input

The `.cntl` and `.strc` files are read into linked lists (`LINKEDLIST$...`). Blocks start with `!NAME` and end with `!END`; a block whose name is changed (e.g. `!ORBPOT_OFF`) is ignored. `READ=F` inside a block also switches the block off. The routines `READIN_*` in `vespaw_ioroutines.f90` transfer the values to the modules.

9.5 Adding a functional

1. Implement energy density and potential in `vespaw_dft.f90` (or a module of its own), register a type name in `READIN_DFTTYPEFROMNAME`.
2. Provide the one-center evaluation (`vespaw_augmentation.f90`), with the full potential, Eq. (2.18).
3. Verify: energy vs finite difference of the potential along a density change; variational optimizer vs damped dynamics (Sec. 6.7); forces and stress vs finite differences.
4. Add a regression case.

9.6 Adding an optimizer

Optimizers are dispatched on `OPTIMIZERTYPE` in `WAVES$ETOT`. A new eigensolver must return orthonormal Ritz vectors and their Ritz values in `THIS%EIGVAL` (used for the Lagrange multipliers of the forces). It should be tested against Davidson on the whole regression suite.

9.7 Regression suite

`regression/run.py` runs all cases (`regression/*/case.json`) or selected ones (`run.py CASE:RUN`). A case has template inputs with `@PLACEHOLDER@` substitutions, several runs (optimizers, cold and seeded starts, multi-stage runs sharing a restart file), checks against reference

values, agreement tests between runs, finite-difference tests of forces and stress, and a list of known issues. The cases cover PBE Si (all optimizers), hot Al (metal, Mermin), Si_8 at Γ, NiO AFM (PBE, PBE+U, PBE0r, PBE-SL0r, SCAN0r, TSCAN), MgO PBE-SL0r, TSCAN Si, spin–orbit Si, a polar slab with dipole correction, space-group symmetry, and forces and stress. Every change to the code should leave the suite passing.

Bibliography

[1] P. E. Blöchl. Projector augmented-wave method. *Phys. Rev. B*, 50:17953–17979, 1994. doi: 10.1103/PhysRevB.50.17953.

[2] M. Born and R. Oppenheimer. Zur quantentheorie der molekeln. *Ann. Phys.*, 389:457, 1927.

[3] M. Born and K. Huang. *Dynamical Theory of Crystal Lattices*. Oxford University Press, Oxford, 1954.

[4] R. P. Feynman. Forces in molecules. *Phys. Rev.*, 56:340, 1939.

[5] A. Szabo and N. S. Ostlund. *Modern Quantum Chemistry*. McGraw-Hill, New York, 1989.

[6] P. Hohenberg and W. Kohn. Inhomogeneous electron gas. *Phys. Rev.*, 136:B864, 1964.

[7] M. Levy. Universal variational functionals of electron densities, first-order density matrices, and natural spin-orbitals and solution of the v-representability problem. *Proc. Natl. Acad. Sci. USA*, 76:6062, 1979.

[8] W. Kohn and L. J. Sham. Self-consistent equations including exchange and correlation effects. *Phys. Rev.*, 140:A1133, 1965.

[9] J. Harris and R. O. Jones. The surface energy of a bounded electron gas. *J. Phys. F*, 4: 1170, 1974.

[10] D. C. Langreth and J. P. Perdew. The exchange-correlation energy of a metallic surface. *Solid State Commun.*, 17:1425, 1975.

[11] O. Gunnarsson and B. I. Lundqvist. Exchange and correlation in atoms, molecules, and solids by the spin-density-functional formalism. *Phys. Rev. B*, 13:4274, 1976.

[12] D.M. Ceperley and B.J. Alder. Ground state of the electron gas by a stochastic method. *Phys. Rev. Lett.*, 45:566, 1980.

[13] J.P. Perdew and A. Zunger. Self interaction correction to density-functional approximations for many-electron systems. *Phys. Rev. B*, 23:5048, 1981.

[14] J. P. Perdew and Y. Wang. Accurate and simple analytic representation of the electron-gas correlation energy. *Phys. Rev. B*, 45:13244, 1992.

[15] D. C. Langreth and M. J. Mehl. Beyond the local-density approximation in calculations of ground-state electronic properties. *Phys. Rev. B*, 28:1809, 1983.

[16] John P. Perdew, Kieron Burke, and Matthias Ernzerhof. Generalized gradient approximation made simple. *Phys. Rev. Lett.*, 77:3865, 1996.

[17] B. Hammer, L.B. Hansen, and J.K. Norskov. Improved adsorption energetics within density-functional theory using revised perdew-burke-ernzerhof functionals. *Phys. Rev. B*, 59:7413, 1999.

[18] J.P. Perdew, J.A. Chevary, S.H. Vosko, K.A. Jackson, M.P. Pederson, D.J. Singh, and C. Fiolhais. Atoms, molecules solids and surfaces: applications of the generalizd gradient approximation for exchange and correlation. *Phys. Rev. B*, 46:6671, 1992.

[19] Y. Zhang and W. Yang. Comment on "generalized gradient approximation made simple". *Phys. Rev. Lett.*, 80:890, 1998.

[20] J. P. Perdew, A. Ruzsinszky, G. I. Csonka, O. A. Vydrov, G. E. Scuseria, L. A. Constantin, X. Zhou, and K. Burke. Restoring the density-gradient expansion for exchange in solids and surfaces. *Phys. Rev. Lett.*, 100:136406, 2008.

[21] A. D. Becke. Density-functional exchange energy with correct asymptotic behavior. *Phys. Rev. A*, 38:3098, 1988.

[22] C. Lee, W. Yang, and R.G. Parr. Development of the colle-salvetti correlation-energy formula into a functional of the electron density. *Phys. Rev. B*, 37:785, 1988.

[23] J. P. Perdew. Density functional approximation for the correlation energy of the inhomogeneous electron gas. *Phys. Rev. B*, 33:8822, 1986.

[24] Jianwei Sun, Adrienn Ruzsinszky, and John P. Perdew. Strongly constrained and appropriately normed semilocal density functional. *Phys. Rev. Lett.*, 115:036402, 2015.

[25] James W. Furness, Aaron D. Kaplan, Jinliang Ning, John P. Perdew, and Jianwei Sun. Accurate and numerically efficient r^{2}SCAN meta-generalized gradient approximation. *J. Phys. Chem. Lett.*, 11:8208, 2020. doi: 10.1021/acs.jpclett.0c02405. URL https://doi.org/10.1021/acs.jpclett.0c02405. Erratum: J. Phys. Chem. Lett. **11**, 9248 (2020).

[26] Daniel Mejía-Rodríguez and S. B. Trickey. Deorbitalization strategies for meta-generalized-gradient-approximation exchange-correlation functionals. *Phys. Rev. A*, 96:052512, 2017.

[27] J. P. Perdew and L. A. Constantin. Laplacian-level density functionals for the kinetic energy density and exchange-correlation energy. *Phys. Rev. B*, 75:155109, 2007.

[28] Daniel Mejía-Rodríguez and S. B. Trickey. Deorbitalized meta-GGA exchange-correlation functionals in solids. *Phys. Rev. B*, 98:115161, 2018.

[29] N. D. Mermin. Thermal properties of the inhomogeneous electron gas. *Phys. Rev.*, 137: A1441, 1965.

[30] H. J. Monkhorst and J. D. Pack. Special points for brillouin-zone integrations. *Phys. Rev. B*, 13:5188, 1976.

[31] M. C. Payne, M. P. Teter, D. C. Allan, T. A. Arias, and J. D. Joannopoulos. Iterative minimization techniques for ab initio total-energy calculations: molecular dynamics and conjugate gradients. *Rev. Mod. Phys.*, 64:1045, 1992.

[32] D. R. Hamann, M. Schlüter, and C. Chiang. Norm-conserving pseudopotentials. *Phys. Rev. Lett.*, 43:1494, 1979.

[33] O. K. Andersen. Linear methods in band theory. *Phys. Rev. B*, 12:3060, 1975.

[34] F. Wooten. *Optical Properties of Solids*. Academic Press, New York, 1972.

[35] R. F. Egerton. Limits to the spatial, energy and momentum resolution of electron energy-loss spectroscopy. *Ultramicroscopy*, 107:575, 2007.

[36] M. P. Seah and W. A. Dench. Quantitative electron spectroscopy of surfaces: A standard data base for electron inelastic mean free paths in solids. *Surf. Interface Anal.*, 1:2, 1979.

[37] R. Laskowski and P. Blaha. Understanding the $L_{2,3}$ x-ray absorption spectra of early 3*d* transition elements. *Phys. Rev. B*, 82:205104, 2010.

[38] I. Davoli, A. Marcelli, A. Bianconi, M. Tomellini, and M. Fanfoni. Multielectron configurations in the x-ray-absorption near-edge structure of NiO at the oxygen K threshold. *Phys. Rev. B*, 33:2979, 1986.

[39] G. Kresse and D. Joubert. From ultrasoft pseudopotentials to the projector augmented-wave method. *Phys. Rev. B*, 59:1758, 1999.

[40] P. E. Blöchl, C. F. J. Walther, and Th. Pruschke. Method to include explicit correlations into density-functional calculations based on density-matrix functional theory. *Phys. Rev. B*, 84:205101, 2011. doi: $10.1103/\mathrm{PhysRevB}.84.205101$.

[41] Mohsen Sotoudeh, Sangeeta Rajpurohit, Peter Blöchl, Daniel Mierwaldt, Jonas Norpoth, Vladimir Roddatis, Stephanie Mildner, Benedikt Ifland, and Christian Jooss. Electronic structure of $Pr_{1-x}Ca_xMnO_3$. *Phys. Rev. B*, 95:235150, 2017. doi: $10.1103/\mathrm{PhysRevB}.95.235150$.

[42] A. D. Becke. A new mixing of hartree–fock and local density-functional theories. *J. Chem. Phys.*, 98:1372, 1993.

[43] John P. Perdew, Matthias Ernzerhof, and Kieron Burke. Rationale for mixing exact exchange with density functional approximations. *J. Chem. Phys.*, 105:9982, 1996.

[44] C. Adamo and V. Barone. Toward reliable density functional methods without adjustable parameters: The PBE0 model. *J. Chem. Phys.*, 110:6158, 1999.

[45] P. J. Stephens, F. J. Devlin, C. F. Chabalowski, and M. J. Frisch. Ab initio calculation of vibrational absorption and circular dichroism spectra using density functional force fields. *J. Phys. Chem.*, 98:11623, 1994.

[46] A. D. Becke. Density-functional thermochemistry. iii. the role of exact exchange. *J. Chem. Phys.*, 98:5648, 1993.

[47] Ole K. Andersen and Ove Jepsen. Explicit, first-principles tight-binding theory. *Phys. Rev. Lett.*, 53:2571, 1984.

[48] P. E. Blöchl and C. Först. Node-less atomic wave functions, Pauli repulsion and systematic projector augmentation. *arXiv:1210.5937*, 2012.

[49] P. Novák, J. Kuneš, L. Chaput, and W. E. Pickett. Exact exchange for correlated electrons. *Phys. Status Solidi B*, 243:563, 2006.

[50] F. Tran, P. Blaha, K. Schwarz, and P. Novák. Hybrid exchange-correlation energy functionals for strongly correlated electrons: Applications to transition-metal monoxides. *Phys. Rev. B*, 74:155108, 2006.

[51] P. Pulay. Convergence acceleration of iterative sequences. the case of scf iteration. *Chem. Phys. Lett.*, 73:393, 1980.

[52] D. G. Anderson. Iterative procedures for nonlinear integral equations. *J. ACM*, 12:547, 1965.

[53] G. P. Kerker. Efficient iteration scheme for self-consistent pseudopotential calculations. *Phys. Rev. B*, 23:3082, 1981.

[54] J. Harris. Simplified method for calculating the energy of weakly interacting fragments. *Phys. Rev. B*, 31:1770, 1985.

[55] W. M. C. Foulkes and R. Haydock. Tight-binding models and density-functional theory. *Phys. Rev. B*, 39:12520, 1989.

[56] X. Gonze. Towards a potential-based conjugate gradient algorithm for order-N self-consistent total energy calculations. *Phys. Rev. B*, 54:4383, 1996.

[57] N. Marzari, D. Vanderbilt, and M. C. Payne. Ensemble density-functional theory for ab initio molecular dynamics of metals and finite-temperature insulators. *Phys. Rev. Lett.*, 79:1337, 1997.

[58] C. Freysoldt, S. Boeck, and J. Neugebauer. Direct minimization technique for metals in density functional theory. *Phys. Rev. B*, 79:241103(R), 2009.

[59] M. P. Teter, M. C. Payne, and D. C. Allan. Solution of schrödinger's equation for large systems. *Phys. Rev. B*, 40:12255, 1989.

[60] M. Sotoudeh. The Kohn-Sham hamiltonian as the variational object. *in preparation*, 2026.

[61] D. M. Wood and A. Zunger. A new method for diagonalising large matrices. *J. Phys. A*, 18: 1343, 1985.

[62] G. Kresse and J. Furthmüller. Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set. *Phys. Rev. B*, 54:11169, 1996.

[63] R. Car and M. Parrinello. Unified approach for molecular dynamics and density-functional theory. *Phys. Rev. Lett.*, 55:2471, 1985.

[64] E. R. Davidson. The iterative calculation of a few of the lowest eigenvalues and corresponding eigenvectors of large real-symmetric matrices. *J. Comput. Phys.*, 17:87, 1975.

[65] O. H. Nielsen and R. M. Martin. Quantum-mechanical theory of stress and force. *Phys. Rev. B*, 32:3780, 1985.

[66] P. E. Blöchl and M. Parrinello. Adiabaticity in first-principles molecular dynamics. *Phys. Rev. B*, 45:9413, 1992.

[67] S. Nosé. A unified formulation of the constant temperature molecular dynamics methods. *J. Chem. Phys.*, 81:511, 1984.

[68] P. E. Blöchl. Second-generation wave-function thermostat for ab initio molecular dynamics. *Phys. Rev. B*, 65:104303, 2002.